

PLASTIBIOFUEL LLC · HASLET, TEXAS

# Post-consumer PET to chemicals

One page, written to be forwarded. Every number on it traces to a signed independent report.

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| <b>122</b><br>MICROMOL/H/MG TURNOVER, IMMOBILIZED | <b>3</b><br>MILESTONES CLOSED EARLY | <b>1</b><br>PROVISIONAL PATENT |
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| <b>ENTITY</b><br>PlastiBioFuel LLC. UEI KQAWZ54RDUM8, CAGE 175D6. Service-disabled veteran-owned small business, certified through SBA's Veteran Small Business Certification program. | <b>NAICS</b><br>325199 other basic organic chemical manufacturing; 325193 ethyl alcohol manufacturing; 541715 research and development. |
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| <b>WHAT WE DO</b><br>A PET hydrolase immobilized in a covalent organic framework depolymerizes post-consumer PET. A continuous spiral-flow reactor, designed in simulation, is the intended production form. | <b>RESEARCH PARTNER</b><br>University of North Texas, Prof. Shengqian Ma, under Sponsored Research Agreement SRA1016. Framework and immobilization chemistry. |
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| <b>INTELLECTUAL PROPERTY</b><br>USPTO provisional application 63/848,456, filed July 2025. Framework immobilization, spiral-flow reactor geometry, and the integrated process. |
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| <b>ESTABLISHED BY INDEPENDENT REPORT</b> | UNT Final Project Report, Phase I, signed 2026-08-07 |
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| <b>Depolymerization confirmed</b> | Free FAST-PETase breaks real post-consumer water bottle PET down to terephthalic acid, confirmed by HPLC against standards.               |
| <b>Immobilization confirmed</b>   | The enzyme loads into a TAPB-BPDA-COF with no protein detectable in the supernatant at 280 nm.                                            |
| <b>Activity retained</b>          | Immobilized turnover measured 122 micromol per hour per milligram at 24 hours, against a 246 to 348 literature range for the free enzyme. |

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| <b>Bottleneck identified</b> | The laboratory attributes that gap to mass transfer down the framework channel rather than to loss of enzyme activity. Transport is what reactor geometry decides. |
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| <b>Schedule</b> | Three milestones completed ahead of schedule. |
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### NOT YET ESTABLISHED, STATED PLAINLY

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| <b>No reactor built</b> | The spiral-flow reactor exists as a validated model. The report lists vessel shape and size as future work. |
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| <b>No ethanol produced</b> | The route from monomer to fuel-grade ethanol is the objective and has not been demonstrated. |
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| <b>No offtake executed</b> | Commercial offtake is in discussion. There is no signed letter of intent and no executed agreement. |
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| <b>Fuel credits unqualified</b> | Any renewable fuel credit eligibility remains subject to final pathway qualification. |
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#### WHAT WE ARE LOOKING FOR

Partners and programs in chemicals from waste feedstocks, pre-pilot reactor capacity, and downstream separations. First-time federal applicant, prepared to work as prime or subrecipient.

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UEI KQAWZ54RDUM8 · CAGE 175D6